target tracking, and data collection across military and civilian sectors. The period also saw the gradual expansion of drone technology into scientific research, environmental monitoring, and commercial applications, laying the groundwork for the intelligent, AI-driven UAVs that followed.

In essence, the second generation of drones represents the technological and operational evolution from purely manual control systems to remotely guided, sensor-assisted, and GPS-enabled platforms, establishing drones as practical tools for both defense and non-military missions.

3.1.2 Basic characteristics of second generation of drones

The basic characteristics of second-generation drones reflect a major leap in technology, design, and operational flexibility. The following features distinguish them from the simpler, mechanically operated first-generation UAVs:

- *Advanced radio control systems*: Equipped with multi-channel, frequency-modulated (FM) transmitters and receivers, enabling precise and interference-resistant control over long distances.
- *GPS-based navigation*: Early integration of the global positioning system (GPS) allowed drones to follow pre-defined routes, conduct way-point navigation, and return to launch points autonomously.
- *Digital communication and telemetry*: Real-time transmission of flight data, imagery, and sensor feedback to ground control stations enhanced situational awareness and mission efficiency.
- *Flight stabilization and onboard electronics*: Inclusion of inertial measurement units (IMUs), gyroscopes, and flight controllers improved aerial stability and allowed semi-autonomous flight under variable weather conditions.